A CONCRETE QUESTION ABOUT FINITE OBSERVERS

Finite Directional Records as a Concrete Observer Model

Can one localized system acquire, retain, and expose a relational orientation record while its information, support, energy cost, and gravitational footprint are controlled in one semiclassical regime?

The observer-level question

Quantum-gravity observer rules give finite operational content to an observer, while de Sitter algebras include it in the construction [1]-[4]. We ask whether a localized register can make assumptions auditable by storing another system's relative orientation for later readout.

The aim is not to identify a rotor with an observer, but to expose the needed information, dynamics, localization, and gravity.

Why direction is useful

Finite spatial reference frames and their degradation are established [5]-[7]. We use a wholly relational task: if `O` is the register and `D` the target, the unknown is their alignment, not orientation relative to coordinates:

```
g_rel = g_O^(-1) g_D in SO(3).                    (1)
```

Performance is the Haar-prior global risk for estimating `g_rel`. It probes the full rotational orbit, unlike local sensitivity or Hilbert-space size alone.

The record variable

For the covariant orientation ensemble, we begin with the relative entropy of rotational asymmetry:

```
S_dir := D(rho_O || G_SO(3)[rho_O])
= S(G_SO(3)[rho_O]) - S(rho_O).                   (2)
```

`S_dir` is relative entropy of rotational asymmetry. It is not identified with classical memory dimension, thermodynamic entropy, Hilbert-space dimension, Casimir, quantum Fisher information, or Harlow's `S_Ob`. Whether comparison with `S_Ob` is meaningful is review question 1.

Lifetime is operational: after time `T`, a permitted relational readout must still attain the declared risk. This is classical-record stability, not generic phase coherence.

The proposed theorem chain

Fix a class `C_dir` of localized register-target models, states, environments, actions, and preparation/storage/readout protocols. The intended theorem quantifies over this class:

```
R_rel(T) >= F_record(S_dir, degradation, channel
error)
S_dir <= F_capacity(energy, support, gravity;
model data)
=> Phi(R_rel(T), T, support, gravity; model data)
>= 0.                                             (3)
```

Rigorous arrows exist on named domains: global risk requires rotational resources; confined orbital matter converts them into support and excitation-energy cost; a stipulated isotropic heat channel gives a conditional degradation law.

The primary bottleneck

The common-action physical record channel is **open**. One local action must derive acquisition, storage degradation, and readout; bound the microscopic KMS-to-effective-channel error; and enter the same stress-energy ledger. Existing de Sitter master equations specify the needed approximation, not that it follows from thermality [10].

What gravity adds, and what the sprint decided

Exact fixed-de Sitter maps define a Weyl witness [11]. Its estimator is negative, but the bounded sprint's full interval contains zero because the primal certificate is about an order of magnitude too broad. Paper R stops as a viability test: a proof limitation, not physical cancellation. The witness remains suggestive, not a theorem or capacity bound.

A supported massive Skyrmion supplies one certified matter candidate [12]. It only makes the supported-profile subproblem nonempty; the architecture must survive rejection of this realization.

Why request feedback now

The STOP makes the fork explicit. We ask whether finite directional records are useful, how `S_dir` relates to `S_Ob`, what gravitational quantity constrains the register, and whether a one-action model helps without a closed-universe path integral. We seek redirection, not endorsement or a publication decision.

Claim boundary: `S_dir != S_Ob`; record stability is not quantum coherence; measurement back-action is not gravitational backreaction; fixed-background response is not self-consistent gravity.

Finite directional records as an observer model

Dependency map: rigorous ingredients, conditional bridges, and the theorem still to be built

Task: infer the relative SO(3) alignment of two localized physical systems - never an absolute direction.

STATUS [PROVED] theorem on a declared domain · [CONDITIONAL] model-dependent or floating evidence · [OPEN]

OBSERVER-REGISTER CHAIN

Relational global-risk task [PROVED]

Haar-prior full-frame alignment of register O relative to target D.

[PROVED] global information-risk theorem

Required rotational asymmetry S_dir [PROVED]

Relative entropy of SO(3) asymmetry for the declared covariant task.

[PROVED] sector and matter-capacity lemmas

Representation and energy cost [PROVED]

Spin-sector Gibbs bounds and localization-energy bounds in named orbital or supported-collective matter classes.

[CONDITIONAL] support and gravity dictionary

Localization and gravity budgets [CONDITIONAL]

Support and energy are controlled in named models; the invariant that should cap observer capacity is not yet identified.

[CONDITIONAL] witness branch

[OPEN] common-action record channel

Record lifetime and readout error [CONDITIONAL]

Exact degradation and readout implications exist for stipulated effective channels; action-derived preparation, storage, and readout remain open.

[OPEN] joint elimination and common window

Proposed localized-observer tradeoff [OPEN]

One action, one state family, and one semiclassical regime jointly control risk, lifetime, support, energy, and gravitational backreaction.

REALIZATION SUPPORT

Paper A: certified profile [PROVED]

Hard-wall, fixed-de Sitter: existence, local uniqueness, monotonicity, finite inertia.

Candidate register [CONDITIONAL] Not rotating or self-gravitating.

[CONDITIONAL] candidate source

Paper R: STOP diagnostic [PROVED]

Bounded fixed-background sprint.

[CONDITIONAL] Signed estimator is stably negative.

[PROVED] Full rigorous response interval contains zero.

[OPEN] B_W zero exclusion

INCONCLUSIVE STOP; not promoted and not a physical no-go.

[OPEN] relevant gravity cap?

NON-NEGOTIABLE CLAIM BOUNDARIES

[PROVED] Relational, not absolute

The datum is alignment between two physical systems, not orientation relative to coordinates.

[OPEN] S_dir != S_Ob

No observer-entropy dictionary is proved.

[CONDITIONAL] Fixed background

B_W is a candidate witness, not the capacity bound and not self-consistent gravity.

[OPEN] Common-action channel

Preparation, storage, and readout are not yet derived together.

THE PURPOSE OF THIS PACKET

Where the observer model should connect to quantum gravity

The response sprint did not certify zero exclusion. These questions decide whether a different bridge is worth building.

1. Is `S_dir`, used here as an operational `SO(3)`-asymmetry resource for a covariant alignment task, meaningfully comparable to the observer entropy `S_Ob` in the observer rule, or are these fundamentally different resources?

2. Which gauge-invariant quantity should replace or refine the fixed-background Weyl witness as a capacity constraint in a de Sitter patch: collapse, horizon or quantum-extremal-surface displacement, or another invariant?

3. Would a one-action semiclassical model deriving relational record formation, lifetime, localization, readout, and gravitational footprint clarify the observer rule, or is a closed-universe gravitational path integral essential from the outset?

Feedback target. We are not asking whether the present realization is complete. We are asking whether this is the right operational bridge to build, which quantity should replace the provisional gravitational witness if needed, and which observer-level identification should remain deliberately absent.

Auditable Status of the Directional-Record Program

[PROVED] means an analytic or computer-assisted theorem on the stated domain. **[CONDITIONAL]** means an exact implication under a stipulated effective model or source-bound floating evidence. **[OPEN]** means a missing implication, norm estimate, or physical identification. A composite claim inherits the least-complete label of its ingredients.

A. Relational orientation as the operational task

[PROVED] Let the unknown relative orientation `g in SO(3)` have normalized Haar prior, let an arbitrary POVM return `g_hat`, and use full-frame chordal cost `sin^2(theta(g_hat^-1 g)/2)`. The reference representation may contain arbitrary rotation-trivial multiplicities. For every normal state and POVM, including mixed states and invariant ancillas,

```
R_ref >= 1/(16 <J^2>+8),
R_ref >= c_SO3 exp[-2 S_dir/3],   c_SO3=6/(e pi^(5/3)),
S_dir=A_SO3(rho)=D(rho || G_SO3(rho)).                                (A.1)
```

The first bound follows from the spin-1 fusion matrix and a discrete Hardy inequality. The second follows from relative-entropy data processing for the covariant orbit ensemble and a global rate-distortion bound. Neither proof assumes a locally unbiased estimator, Cramer-Rao regime, purity, polarization, fixed irrep, or finite-dimensional multiplicity. This is Research Theorem W3 in `THEOREMS.md` and `docs/global_so3_reference_risk.md`.

[PROVED] The datum is the alignment of two physical frames, never orientation relative to coordinates. The loss is global over the group, not a local-sensitivity proxy. For strict integer-spin support through `j=J`, `R_ref>=sin^2[pi/(2J+3)]`. The fermionic Finkelstein-Rubinstein sector has its separately stated projective bound.

[PROVED] Here `S_dir` is only relative entropy of rotational asymmetry. Equation (A.1) gives the necessary allocation `S_dir>=(3/2)log(c_SO3/R_ref)` where positive. It does not establish accessible classical memory, sufficiency, or a relation to `S_Ob`.

[CONDITIONAL] If the reference alone undergoes stipulated isotropic rotational heat diffusion with exposure `Gamma(T)=integral_0^T gamma(t)dt`, the spin-1 score is attenuated exactly:

```
R_ref(T) >= (3/4)[1-exp(-2 Gamma)]
+ exp(-2 Gamma)/(16 <J^2>+8).                                         (A.2)
```

This is a finite-time record-risk theorem for that effective channel, not preservation of arbitrary quantum coherence. **[OPEN]** No one-action protocol yet controls preparation, storage, readout, microscopic-to-effective channel error, support stress, and backreaction on one nonempty open parameter set.

B. Capacity from localization and energy

[PROVED] For finitely many spinless nonrelativistic particles of total mass `M`, hard support `|x_i|<=a`, and a ground-subtracted excitation Hamiltonian satisfying the quadratic-form inequality `H_ex>=sum_i p_i^2/(2m_i)`, every form-domain state obeys

```
<L^2> <= 2 M a^2 E_ex,
R_ref >= 1/(32 M a^2 E_ex+8).                                    (B.1)
```

The result includes mixed, zero-mean, and rare-tail states with rotation-trivial internal labels. It excludes intrinsic spin, relativistic or massless fields, soft localization, negative interaction energy, and nontrivially rotating light species. A separate compactness substitution is a declared proxy, not a general-relativistic body theorem. See W3a and `docs/localized_orbital_reference.md`.

[PROVED] More generally, sector floors `epsilon_j` yield `S_dir<=beta E+log Z_H(beta)` only when `Z_H(beta)=sum_j(2j+1)^2 exp(-beta epsilon_j)` is finite. Covariance plus finite mean energy is insufficient; a bounded invariant counterexample permits increasingly accurate frame states at uniformly bounded energy. A physical theorem must derive growing sector floors from its matter model.

[PROVED] In the hard-supported massive-Skyrme hedgehog collective family on fixed de Sitter, with support radius `a` and minimum support lapse `N_w>0`, the static energy and collective inertia obey

```
I[F] <= 4 M[F] a^2/(3N_w),
E_j >= sqrt[3N_w j(j+1)/2]/a.                                    (B.2)
```

This produces a linear large-spin floor and finite integer or projective rotational partition function. It is uniform over radial relaxation inside the adiabatic collective family (W3b).

[CONDITIONAL] The Skyrmion is therefore a concrete candidate for the capacity premise, not a complete rotating-field observer. Collective-band completeness, noncollective modes, projection error, right/isospin access, and the capacity of a marked rotating wall remain uncontrolled.

C. Certified supported profile

[PROVED] At the declared fixed-background hard-wall parameters, AU.1 is an exact-rational computer-assisted local existence theorem. It proves one solution in a certified Newton ball, strict monotonicity, a negative wall slope, and finite positive rotor inertia. The source artifact is `experiments/skyrmion_newton_reduced_hessian_rounded_exact_certificate.json`, SHA256 prefix `c4c95db47470`.

> Digest strings on pages 5-7 show the first 12 hexadecimal characters. Full SHA256 values and exact artifact paths are indexed in `claim_evidence_ledger.md`.

[PROVED] This is local uniqueness on fixed pure de Sitter with a prescribed Dirichlet wall. It is not global uniqueness, dynamical stability, a rotating solution, arbitrary wall dynamics, or Einstein-Skyrme backreaction. **[CONDITIONAL]** It establishes that the candidate used in later calculations is more than a shooting profile; the wall remains a physical modeling input, not a harmless regulator.

D. Conserved source, master field, and electric Weyl

[PROVED] A static quadrupolar source cannot be specified by energy density alone. W3n gives the exact static even-parity bulk conservation equations and wall tractions. W3o shows that the undeformed rigid rotational Skyrmion stress fails this gate. This rejects the rigid source, not rotating Skyrmions: the centrifugal matter deformation and moving support must be included.

[PROVED] For the declared fixed-background matter-plus-moving-membrane weak form, W3z.15-W3z.16 establish a closed positive Friedrichs operator with `||A^-1||<=100`, nonzero regular forcing, and a unique nonzero weak centrifugal deformation. The source artifacts have SHA256 prefixes `4a4e3ecd48a2` and `9edc2bc47953`. Nonzero matter deformation alone does not imply nonzero exterior response.

[CONDITIONAL] Exact Hilbert-variation and shell identities, with source-bound mesh refinement, support a completed bulk-plus-moving-membrane stress at the default point. The analytic factorization is exact; closure of the selected branch is floating rather than interval certified. The safe wording is "source-bound floating conservation audit," not an unqualified certified source.

[PROVED] Once a valid conserved stress is supplied, the source, propagation, and readout maps are exact in one frozen convention. On fixed pure de Sitter, for the static `ell=2` normalization, regular center behavior, and no-log horizon condition,

```
A_2 Psi = F,   A_2=-d/dr[(1-r^2/R^2)d/dr]+6/r^2,
F=8piG[-r^2 N rho'/6+r(1+4r^2/R^2)rho/6
-r p_r/2-j+2r pi],   delta E_rr=-6 Psi(r)Y_2/r^3.                          (D.1)
```

The Friedrichs operator has a positive exact Green kernel and `A_2>=6/R^2`. W3t, W3v, and W3x prove the stress-to-master identity, ideal-shell transmission, and exterior master-to-electric-Weyl reconstruction. Their artifact prefixes are `3be17612134b`, `c83536779d2a`, and `3025d65fe825`. Exact maps do not certify a nonzero completed input amplitude.

[CONDITIONAL] The completed signed estimator is stably negative, `J_hat in [-0.0030795543,-0.0025529313]`, after including origin, correlated positive-radius, and compatible weak-wall terms. This is encouraging design evidence, not the response theorem: the rigorous residual-product interval on page 7 contains zero.

Witness boundary. If nonzero, the proposed `B_W` would witness non-invisibility on a fixed background. It is not the observer-capacity bound, a self-consistent geometry, or a substitute for collapse, horizon/QES displacement, or full junction data.

E. Why Paper R stopped

[PROVED] The certification architecture is exact. Outside the compact source, the fixed-background response factorizes into one amplitude `A_ext=J_rigid+B(y)`. If the symmetric weak form obeys `q(y,v)=ell(v)` and the adjoint obeys `q(v,z)=B(v)`, arbitrary conforming trials satisfy

```
J_hat=J_rigid+B(y_h)+R_y(z_h),
|A_ext-J_hat| <= ||R_y||_(q*) ||R_z||_(q*).                                    (E.1)
```

Zero exclusion requires the distance of the corrected-estimator interval from zero to exceed the product of the full primal and adjoint dual-residual bounds. This is W3z.17.

[PROVED] The bounded sprint completed every origin, bulk, and wall term. The all-strong primal and weak completed-square adjoint certificates give

```
delta_y < 0.785351351663998829,   delta_z < 0.030892717992632714,
delta_y delta_z < 0.024261637832088839,
A_ext in [-0.027341192151999246,0.021708706437937767].                          (E.2)
```

The full interval contains zero. Under the predeclared bounded-sprint rule, the result is **INCONCLUSIVE STOP**: no nonzero exterior-Weyl theorem or equal-energy state-discrimination corollary is certified. The decision artifact prefix is `bcbb4a1af96b`.

[PROVED] Holding the estimator and adjoint bound fixed, zero exclusion would require `delta_y<0.082638613888227453`, about a `9.50x` reduction from the present primal norm. The dominant primal cell is `[1/2,11/16]`; the global coercivity floor is not the main loss.

[OPEN] Paper R is frozen as a viability memo and not promoted as a manuscript. A future attempt needs a materially different proof object, such as a better conforming primal trial, a certified Riesz solve, or a structural decomposition. More general subdivision of the present representation is deferred.

> **Interpretation boundary.** The negative estimator is suggestive; the zero-containing theorem interval is binding. This certifies failure of the present proof representation, not physical cancellation or a no-go theorem.

F. External claim boundary

[PROVED] The defensible mathematical core is: global relational `SO(3)` risk requires asymmetry and rotational resources; named localized matter classes convert rotational capacity into energy/support cost; one hard-wall profile and its fixed-background centrifugal inverse exist; and a valid conserved quadrupolar source has exact master, transmission, and Weyl maps.

[CONDITIONAL] The defensible physical construction is narrower: a hard-supported Skyrmion is a plausible semiclassical realization, a stipulated heat channel gives an exact degradation law, and the signed fixed-background estimator favors a nonzero exterior response. The completed rigorous interval does not certify that response.

[OPEN] The project has not identified `S_dir` with `S_Ob`; derived acquisition-through-readout from one action; proved a common parameter window; certified `|B_W|>0`; completed tensorial Israel matching; solved a rotating self-gravitating Einstein-Skyrme-de Sitter geometry; or shown that exterior Weyl response caps observer information. A closed-universe path integral is outside the present claims.

[OPEN] Novelty, priority, venue fit, and publishability are not mathematical outputs of the repository. The separate literature matrix gives the primary-source comparison. Search absence is not used as proof of priority.

G. Reproducibility map

Status	Result	Primary repository pointer
[PROVED]	Global risk/asymmetry/Casimir bounds	`THEOREMS.md` W3; `docs/global_so3_reference_risk.md`; focused tests
[PROVED]	Confined orbital capacity	W3a; `docs/localized_orbital_reference.md`; focused tests
[CONDITIONAL]	Heat-capacity composition	W3a.1; `experiments/universal_observer_tradeoff_certificate.json`
[PROVED]	Supported collective sector floor	W3b; `docs/supported_skyrmion_collective_spectral_floor.md`
[PROVED]	Validated hard-wall profile	AU.1; exact Newton certificate listed on page 5
[PROVED]	Fixed-background source/master/Weyl maps	W3t, W3v, W3x; artifact hashes on page 6
[PROVED]	Paper R viability decision	`docs/paper_r_viability_decision.md`; decision hash on page 7
[CONDITIONAL]	Signed nonzero response evidence	Corrected estimator on page 6; the full interval contains zero
[OPEN]	Nonzero Weyl theorem	Needs a new primal proof object with about a 9.50-fold norm improvement

[PROVED] The complete wording guardrail, exact artifact index, and focused verification record are in `docs/harlow_review_packet/claim_evidence_ledger.md`. The full source comparison and its unresolved search limits are in `literature_matrix.md` and `source_verification_log.md`.

> **Adversarial distinctions.** Absolute orientation versus relational alignment; quantum coherence versus classical record stability; `S_dir` versus `S_Ob`; measurement back-action versus gravitational backreaction; and fixed-background response versus self-consistent gravity are audited separately.

SHORT BIBLIOGRAPHY

Conceptual Neighbors

These twelve primary sources anchor the review question. The separate literature matrix audits 33 sources across observer rules, finite reference frames, clocks, de Sitter open systems, gauge-invariant response, Skyrmions, and rigorous numerics.

1. D. Harlow, M. Usatyuk, and Y. Zhao, "Quantum mechanics and observers for gravity in a closed universe," arXiv:2501.02359 (2025).
2. D. Harlow, "Observers, alpha-parameters, and the Hartle-Hawking state," arXiv:2602.03835 (2026).
3. V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An Algebra of Observables for de Sitter Space," arXiv:2206.10780 (2022).
4. J. De Vuyst, S. Eccles, P. A. Hoehn, and J. Kirklin, "Gravitational entropy is observer-dependent," arXiv:2405.00114 (2024).
5. E. Bagan, M. Baig, and R. Munoz-Tapia, "Aligning Reference Frames Using Quantum States," arXiv:quant-ph/0106014 (2001).
6. S. D. Bartlett, T. Rudolph, R. W. Spekkens, and P. S. Turner, "Degradation of a quantum reference frame," arXiv:quant-ph/0602069 (2006).
7. G. Gour, I. Marvian, and R. W. Spekkens, "Measuring the quality of a quantum reference frame: the relative entropy of frameness," arXiv:0901.0943 (2009).
8. B. A. Costa and Y. E. Sanchez Sanchez, "An Upper Bound for the Mass of Microscopic Clocks," arXiv:2602.24177 (2026).
9. L.-Q. Chen and F. Giacomini, "Quantum Reference Fields Transformations in Linearized Quantum Gravity," arXiv:2606.09344 (2026).
10. M. Fukuma, Y. Sakatani, and S. Sugishita, "Master equation for the Unruh-DeWitt detector and the universal relaxation time in de Sitter space," arXiv:1305.0256 (2014).
11. H. Kodama and A. Ishibashi, "A master equation for gravitational perturbations of maximally symmetric black holes in higher dimensions," arXiv:hep-th/0305147 (2003).
12. Y. Brihaye and T. Delsate, "Skyrmion and Skyrme-Black holes in de Sitter spacetime," arXiv:hep-th/0512339 (2006).

Literature boundary. Existing work already covers finite observer entropy, observer-dependent gravitational algebras, degrading quantum reference frames, physical clock limits, material reference fields, de Sitter open-system dynamics, gauge-invariant response, and curved/cavity Skyrmions. The proposed distinction is their quantitative conjunction in one narrow relational `SO(3)` record model. No priority claim is made. Journal databases and terminology outside the targeted search were not exhausted.